CSA15 Cloud Computing and Big Data Analytics

Capstone Cloud Technical Presentation

SkillHire Cloud Resume-to-Job Matcher Using NLP Skill Gap Intelligence

Presentation Focus

Present your capstone as a cloud-backed product. Explain the problem, architecture, service choices, evidence, and CO1 learning clearly.

Presenter Details

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Class / Section / Slot:CSA1512
Capstone Title: SkillHire Cloud Resume-to-Job Matcher Using NLP Skill Gap Intelligence.
Selected SDG Goal(s):
Cloud Platform / Tools:Azure

Guidance - delete before presentation: Fill presenter details, then delete this orange guidance tab from every slide before final submission.

Problem Context and Stakeholder Mapping

Our capstone begins with a real problem to solve

Problem Statement

- Job seekers often apply for jobs without knowing whether their skills match the job requirements.
- Manually comparing resumes with job descriptions is time-consuming.
- Traditional resume screening may not clearly identify missing skills.

Target Users

- Job seekers
- Students and fresh graduates
- Working professionals
- HR teams

SDG Fit

- **SDG 4:** Supports continuous learning and skill development.
- **SDG 8:** Helps people find suitable employment opportunities.

Product Boundary

- Resume upload.
- Resume text extraction.
- Job description input.
- NLP-based skill extraction.

Cloud Adoption Rationale and Concept Map

The problem needs cloud support to become a usable product

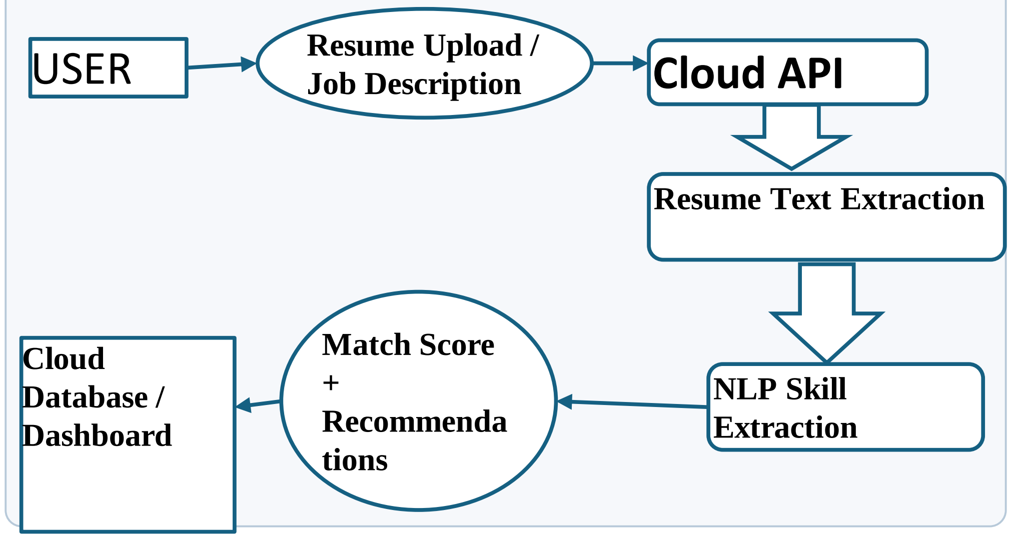
Cloud Definition

In SkillHire, cloud technology allows users to upload resumes and receive job-matching results from anywhere.

Cloud Characteristics

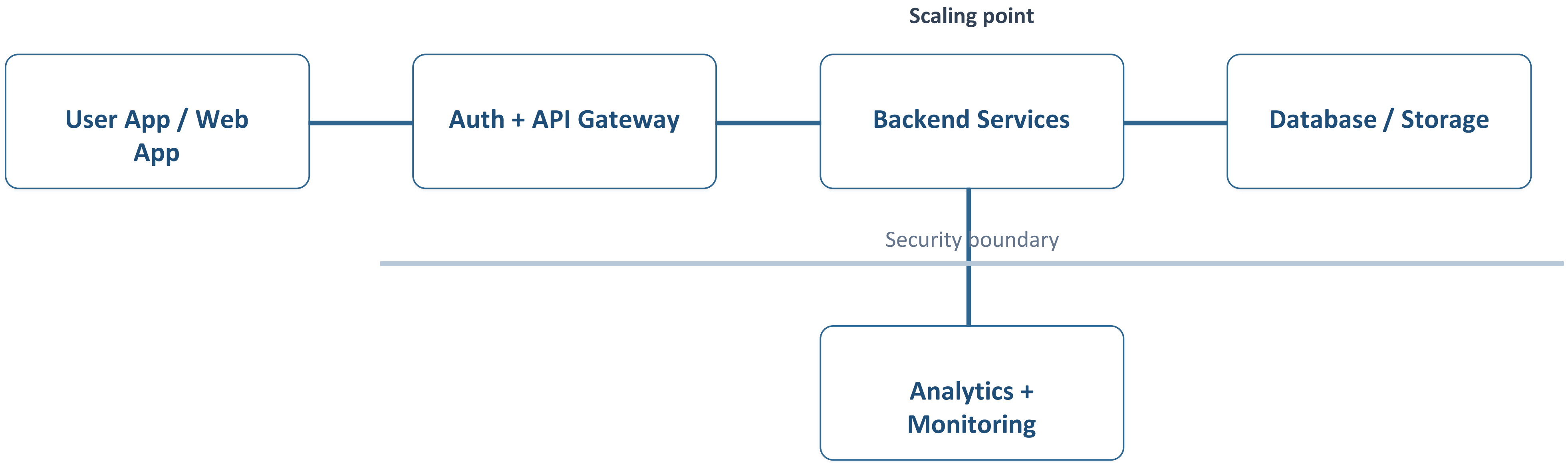
- On-demand resource availability.
- Scalability.
- Elasticity.
- Resource pooling.

Cloud Concept Map



Capstone Cloud Architecture

Our cloud architecture connects the user action to services



Request Path

User → Web App → API → Backend → NLP Engine → Matching Engine → Database → Result

Cloud Service Model Responsibility Matrix

Each service model defines what we control and what cloud manages

Service area	Model	Provider manages	Student team manages	Why this fits
Frontend hosting	PaaS / Static Hosting	Servers, infrastructure	Application files	Easy deployment
Backend/API	PaaS	Runtime& infrastructure	API and business logic	Simple scaling
Database	DBaaS	Database infrastructure	Data and schema	Managed storage
Storage	Cloud Storage	Storage infrastructure	Files and access policies	Secure document storage
Monitoring / logs	PaaS / Serverless	Execution environment	NLP model and code	Flexible AI processing
Security / IAM	Managed Security	Security infrastructure	Roles and permissions	Controlled access

Data Flow and Analytics Pipeline

The product data moves from user activity to useful insight

Collect — Ingest — Store — Process — Visualize

Data Requirements

User data, events, logs, files, sensor data, or transactions.

Analytics Requirement

Dashboard, alert, prediction, report, trend, or decision support.

Proof Required

Sample input, sample output, and one insight generated.

Python / AI Module Integration

Python or AI adds intelligence where the product needs it

Python Role

- Resume text extraction
- Natural Language Processing
- Skill identification
- Keyword extraction

Input and Output

Resume document
Job description

Execution Point

- Cloud API endpoint
- Containerized Python application
- Serverless function
- Cloud-hosted NLP service

Product Value

- Saves time for job seekers.
- Provides personalized skill recommendations.
- Improves job matching.

Security, Scalability, Privacy and Cost Design

Security, scale, privacy and cost shape the final design

Security

- User authentication
- Role-based access control
- Secure API endpoints
- Input validation

Privacy

- Resumes contain personal information.
- Restrict access to uploaded resumes.
- Use secure authentication.

Scalability

- Cloud infrastructure can support increasing users.
- Auto-scaling can handle high traffic.
- Serverless services can process requests dynamically.

Cost Awareness

- Use cloud free tiers during development.
- Use serverless services when suitable.
- Optimize database and storage usage.

Implementation Stack and Deployment Evidence

Our implementation stack shows how the design will be built

Mobile / Web Layer

- React.js / HTML / CSS / JavaScript
- Responsive web interface
- Resume upload interface

Cloud Services

- Cloud compute
- Cloud storage
- Cloud database
- Authentication

Evidence

- **Frontend:** React.js
- **Backend:** Python Flask / FastAPI
- **NLP:** NLTK / spaCy / Transformers

Version Control

- GitHub repository
- Cloud deployment screenshot
- Resume upload screenshot
- API testing screenshot

Demo Storyboard and User Journey

A complete user journey proves the product flow

Scene 1

User Action: Creates an account.
System Response: Account is authenticated.
Cloud Component: Authentication service.

Scene 2

User Action: Uploads resume.
System Response: Resume is stored securely.
Cloud Component: Cloud Storage.

Scene 3

User Action: Enters or selects a job description.
System Response: Job requirements are extracted.
Cloud Component: Backend API.

Scene 4

User Action: Starts resume-job matching.
System Response: NLP extracts relevant skills.
Cloud Component: Python NLP service.

Scene 5

User Action: Views matching results.
System Response: System shows matched and missing skills.
Cloud Component: Matching engine + Database.

Scene 6

User Action: Views improvement suggestions.
System Response: System recommends skills to learn.
Cloud Component: Analytics and recommendation module.

Evaluation Readiness Checklist

Our evidence confirms that the presentation is ready

- [ ] Cloud ecosystem and service model explained
- [ ] Elasticity and scalability decisions are visible
- [ ] REST/SOAP/API role connected to capstone
- [ ] GitHub and cloud evidence are inspectable
- [ ] Architecture has compute, data, storage, security, monitoring
- [ ] References and integrity statement are included

Evaluator Focus:-
The project demonstrates cloud design reasoning, appropriate service selection, NLP integration, scalability, security, and practical deployment evidence.

Conclusion, Limitations and Future Scope

We close with learning, limitations and next steps

Learning Outcome

- Learned how cloud computing supports scalable applications.
- Understood the integration of NLP with cloud services.
- Learned how APIs connect frontend, backend, and AI modules.

Limitations

- Resume formats may vary significantly.
- NLP models may not identify every skill correctly.
- Matching accuracy depends on the quality of resumes and job descriptions.

Future Improvement

- Add advanced Transformer-based NLP models.
- Support multiple languages.
- Add personalized learning-path recommendations.
- Integrate real-time job portals.

Integrity Statement

- All external sources, datasets, tools, libraries, and AI assistance used during the development of the project will be appropriately acknowledged.